

MATH 3551 A — Abstract Algebra I

Practice Final Examination

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For practice — the final is Thursday, December 17, 2026 • Time limit: 2 hours 45 minutes •
Total: 200 points

NAME (please print): _____

Directions. This exam is **closed book**: no textbook, no notes, no calculators, no phones, no other devices, and no assistance of any kind. Write your answers on the exam itself; use the backs of pages if you need more room, and say clearly when you have done so. *Show your work and name the theorems you use* — “by Lagrange’s Theorem, $|H|$ divides 24” earns nearly all of the credit, an unjustified correct number earns little. Point values are marked on each problem. Good luck.

Problem 1. (60 points) **True or false; prove or disprove.**

Circle **T** or **F**, and then give a one-line reason. Three points each.

- (a) **T** **F** There is an element of order 12 in the symmetric group S_7 .
- (b) **T** **F** Every group homomorphism $\mathbb{Z}/12\mathbb{Z} \rightarrow \mathbb{Z}/8\mathbb{Z}$ is trivial.
- (c) **T** **F** The group $(\mathbb{Z}/18\mathbb{Z})^\times$ is cyclic.
- (d) **T** **F** If $H \leq G$ is a subgroup whose index is the smallest prime dividing $|G|$, then $H \trianglelefteq G$.
- (e) **T** **F** Every group of order 105 is abelian.
- (f) **T** **F** The number of 6-cycles in S_8 is 3360.
- (g) **T** **F** $\text{Aut}(\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}) \cong S_3$.
- (h) **T** **F** A_5 contains a subgroup of order 10.
- (i) **T** **F** Every group of order 12 is nilpotent.
- (j) **T** **F** There is a simple group of order 36.
- (k) **T** **F** The quaternion group Q_8 is nilpotent.
- (l) **T** **F** The alternating group A_4 is nilpotent.

- (m) **T** **F** There are exactly five groups of order 8 up to isomorphism.
- (n) **T** **F** Every group of order p^3 , p prime, is abelian.
- (o) **T** **F** There are exactly four groups of order 70 up to isomorphism.
- (p) **T** **F** If G is nonabelian then $G/Z(G)$ is nonabelian.
- (q) **T** **F** The commutator subgroup of D_8 is $\langle r^2 \rangle$.
- (r) **T** **F** For $n \geq 5$ the group A_n is perfect, i.e. $A'_n = A_n$.
- (s) **T** **F** Every quotient group of a nilpotent group is nilpotent.
- (t) **T** **F** There are exactly six abelian groups of order 2^5 up to isomorphism.

Problem 2. (20 points) **A permutation.**

Let $\tau = (1\ 2\ 4\ 7)(3\ 5)(6\ 8\ 9) \in S_9$.

- (a) (3 points) Determine $|\tau|$.
- (b) (4 points) Give the cycle decomposition of τ^{100} .
- (c) (3 points) Is τ even or odd? Justify.
- (d) (5 points) Is τ conjugate in S_9 to $\rho = (1\ 2\ 3\ 4)(5\ 6\ 7)(8\ 9)$? If so, exhibit an explicit $\pi \in S_9$ with $\pi\tau\pi^{-1} = \rho$.
- (e) (5 points) Determine $|C_{S_9}(\tau)|$.

Problem 3. (20 points) **A group of order 204.**

Let G be a group of order $|G| = 204 = 2^2 \cdot 3 \cdot 17$.

- (a) (5 points) Prove that G has a normal Sylow 17-subgroup P .
- (b) (5 points) Prove that G has a subgroup of order 51.
- (c) (7 points) Prove that G is solvable. (*You may want to show first that every group of order 12 has a normal Sylow subgroup.*)
- (d) (3 points) How many elements of order 17 does G have?

Problem 4. (20 points) **The groups of order 55.**

- (a) (4 points) Prove that a group G of order 55 has a normal Sylow 11-subgroup P .
- (b) (4 points) Prove that $G \cong P \rtimes_{\varphi} Q$ for a Sylow 5-subgroup Q and a homomorphism $\varphi: Q \rightarrow \text{Aut}(P)$.
- (c) (8 points) Determine all such φ and conclude that there are exactly two groups of order 55 up to isomorphism.
- (d) (4 points) For the nonabelian group of order 55, determine $Z(G)$ and n_5 .

Problem 5. (20 points) **Abelian groups.**

Let $A = \mathbb{Z}_{36} \times \mathbb{Z}_{50} \times \mathbb{Z}_8$.

- (a) (8 points) Determine $|A|$ in prime-power form, the elementary divisors of A , and the invariant factors of A .
- (b) (4 points) How many abelian groups are there of order $|A|$? (Do not list them.)
- (c) (4 points) What is the largest order of an element of A ? Does A have an element of order 200? Of order 16?
- (d) (4 points) Prove that a finite abelian group is cyclic if and only if all of its Sylow subgroups are cyclic.

Problem 6. (20 points) **Solvable and nilpotent.**

- (a) (5 points) Compute the lower central series of D_8 and give the nilpotence class of D_8 .
- (b) (5 points) Compute the derived series of A_4 and conclude that A_4 is solvable.
- (c) (5 points) Prove that D_{2n} is solvable for every $n \geq 3$.
- (d) (5 points) Prove that A_4 is not nilpotent.

Problem 7. (20 points) **The class equation.**

- (a) (4 points) State the class equation of a finite group G , explaining what each term means.
- (b) (5 points) Determine the class equation of D_8 .
- (c) (5 points) Determine the class equation of A_4 .
- (d) (6 points) Determine the class equation of S_4 .

Problem 8. (20 points) **Isomorphic or not.**

- (a) (5 points) Prove that $\mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/10\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/30\mathbb{Z}$.
- (b) (5 points) Prove that D_{24} (the dihedral group of order 24) is not isomorphic to S_4 .
- (c) (6 points) Prove that Q_8 is not the internal semidirect product $H \rtimes K$ of two proper nontrivial subgroups $H, K \leq Q_8$.
- (d) (4 points) Give, with justification, an example of a group G and a normal subgroup $N \trianglelefteq G$ such that $G \not\cong N \times (G/N)$.

End of practice exam. Total: $60 + 7 \times 20 = 200$ points.